Yash Chopra

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Education

B.Tech in Computer Science and Technology, Maharaja Agrasen Institute of Technology
2024 – 2028 · CGPA: 8.67

Experience

SkillifyMe | Full-Stack Developer Intern **2026**

Portfolio/business site for a Haryana-based architectural surface materials company

- Engineered the admin backend for a full-stack site on React/Vite (Vercel) and Node.js/Express/MongoDB (Render), building JWT-based authentication and role-protected admin routes from scratch.
- Built a product catalog system managing 6 material categories with full CRUD admin controls.
- Designed and shipped 9+ REST API endpoints across admin, inquiry, and product management and Conducted a security review of the codebase and closed 2 critical vulnerabilities – an unauthenticated admin-creation endpoint and unprotected data-mutating routes – before production handoff.

Technical Skills

Languages: Python, C++, Java, JavaScript, SQL
Core CS: DSA, OOP, DBMS, Operating Systems (OS), Computer Networks (CN)
Backend: Node.js, Express.js, PostgreSQL, JWT, FastAPI
Frontend & Infrastructure: React, Redux, Git, Docker, AWS

Projects

Scout | *Python, FastAPI, Node.js, Express, PostgreSQL, pgvector, Groq, Tavily* **2026**

AI research assistant: agentic RAG over web sources and uploaded PDFs [GitHub] [Live]

- **Architected a two-service system:** Python/FastAPI AI service plus Node/Express backend (JWT auth, PostgreSQL), where an LLM agent chooses between Tavily web search and PDF search via tool calling and returns deduplicated, source-attributed answers.
- **Built the RAG pipeline:** chunked PDFs, stored FastEmbed embeddings in PostgreSQL with **pgvector**, retrieved by cosine similarity, and re-ranked with a cross-encoder to ground answers in the document; added a document-only mode that disables web search.
- **Developed a 5-stage deep-research workflow** (plan, execute, analyze, synthesize, verify) that produces structured, verification-checked research reports.
- **Hardened LLM tool calling:** JSON-schema argument validation, bounded retries on provider tool-call failures, a 10-call cap per request, and matched assistant/tool message sequencing, backed by **30+ mocked tests**; secured service-to-service calls with a shared secret and document-ownership checks.

CodeLens | *React.js, Node.js, Express, MongoDB, Gemini 2.5 Flash, JWT, Tailwind CSS* **2026**

AI-powered code review and interview preparation platform [GitHub] [Live]

- **Built an AI code reviewer** on Gemini 2.5 Flash that detects bugs, explains issues, generates corrected code, and gives interview-oriented feedback across **4 languages** (JavaScript, Java, C++, Python).
- **Enforced structured output:** designed JSON response schemas so AI reviews render consistently in the UI and are stored in review history.
- **Implemented secure auth** with JWT and **2 login methods** (email/password and Google OAuth), protected routes, and per-user data access.
- **Developed REST APIs** (Node.js/Express, MongoDB) for authentication, code reviews, review history, interview preparation, and user management.

DevTinder | *React.js, Node.js, Express, MongoDB, Cloudinary, JWT, Tailwind CSS* **2026**

Full-stack developer networking platform [GitHub] [Live]

- **Built a REST API with 10+ endpoints** for authentication, profile management, connection requests, and developer feeds, secured with JWT in HTTP-only cookies and protected routes.
- **Implemented a paginated feed** serving 10 profiles per request, excluding the logged-in user and anyone with an existing connection request.

Positions of Responsibility

Web Development Executive
International Organisation of Software Developers (IOSD), MAIT